

Six Birds: No-Go Theorems for Audited Emergence: Supplement

Ioannis Tsiokos 

Version 1: March 23, 2026 · Version 2: October 3, 2026

How to use this supplement

This supplement collects the repo-facing apparatus that accompanies the main theorem paper. The source repository for the paper, supplement, and reproducibility artifacts is available at <https://github.com/ioannist/six-birds-no-go>. It has two purposes. First, it records provenance, evidence bindings, reproducibility commands, and fragility guardrails in one place. Second, it summarizes the distribution of Lean support across the eight theorem fronts without interrupting the mathematical flow of the main text.

The main paper is intended to stand on its own as a theorem paper. The present supplement is therefore organized around audit and provenance questions: where each claim is bound, which frozen assets support it, which reproduce steps build the relevant summaries, and where the formal boundary is explicitly frozen.

1 Provenance narrative

The theorem line is assembled from a fixed project spine. The theorem order and section homes come from `docs/project/theorem_atlas.yaml`. The theorem statements correspond to `docs/project/theorem_statement_sheet.yaml`, whose Lean scopes are recorded in `review/current-theorem-coverage.json`. The notation registry `docs/project/notation_registry.yaml` records the version 1 conventions; where it differs from the main paper (for example in allowing only positive lags or in its handling of support conditions for KL divergence), the main paper is authoritative.

Claim-level bindings are frozen in `docs/project/claim_audit_freeze.yaml` and `docs/project/theorem_evidence_map.yaml`. Asset choices are frozen in `docs/project/paper_asset_freeze.yaml` and `docs/project/caption_registry.yaml`. Readiness and formal-support status were tracked in `docs/project/readiness_checklist.yaml` together with the per-front Lean bridge files under `docs/project/lean_*.yaml`; these are historical records, and the current formal-support status is the coverage receipt described in the formalization support map below.

The supplement tables do not introduce new mathematical claims. The claim, asset, and callout tables in the theorem–evidence map are version 1 records: they document the frozen claim bindings and planned assets of that version, and their figure and table identifiers and status descriptions do not refer to the figures or formal-support status of the version 2 main paper.

2 Theorem–evidence map

This section records the frozen claim bindings, evidence handles, and asset/callout decisions of version 1 of the paper. They are retained as historical provenance: the asset identifiers (FIG- n , TBL- n) refer to the version 1 asset plan rather than to the figures of the current main paper, and the formal-support descriptions they contain are superseded by the formalization support map. The first table is generated from docs/project/theorem_evidence_map.yaml, docs/project/paper_asset_freeze.yaml, and docs/project/caption_registry.yaml. The second table summarizes the frozen callout bindings and their primary sources.

Table 1: Claim-level theorem–evidence bindings.

Theorem	Claim	Primary handle	Evidence artifacts	Asset / callout decision
NG_ARROW_DPI	NG_ARROW_DPI.core	results/T08/arrow_metrics.json	results/T08/arrow_metrics.json; results/T15/initial_distribution_sweep.csv	FIG-2:repo_asset; Figure 2 ([FIG-2]). Hidden-clock DPI / protocol matrix: honest observ...
NG_PROTOCOL_TRAP	NG_PROTOCOL_TRAP.core	results/T15/initial_distribution_sweep.csv	results/T15/initial_distribution_sweep.csv; results/T08/arrow_metrics.json	FIG-2:repo_asset; Figure 2 ([FIG-2]). Hidden-clock DPI / protocol matrix: honest observ...
NG_FORCE_FOREST	NG_FORCE_FOREST.core	results/T14/primitive_coverage.json	results/T14/primitive_coverage.json; results/master/witness_metrics.csv	FIG-3:repo_asset; Figure 3 ([FIG-3]). Graph / affinity contrasts: forest support remove...
NG_FORCE_NULL	NG_FORCE_NULL.core	results/master/witness_metrics.csv	results/master/witness_metrics.csv	FIG-3:repo_asset; Figure 3 ([FIG-3]). Graph / affinity contrasts: forest support remove...
NG_MACRO_CLOSURE_DEFICIT	NG_MACRO_CLOSURE_DEFICIT.core	results/T09/closure_metrics.json	results/T09/closure_metrics.json; results/T09/grid_check.json	FIG-4:repo_asset; Figure 4 ([FIG-4]). Closure front: exact packaged future laws define ...
NG_OBJECT_CONTRACTIVE	NG_OBJECT_CONTRACTIVE.core	results/T10/objecthood_metrics.json	results/T10/objecthood_metrics.json; results/T10/stability_grid.json	FIG-5:repo_asset; Figure 5 ([FIG-5]). Objecthood front: in the admissible total-variati...
NG_LADDER_IDEM	NG_LADDER_IDEM.core	results/T11/no_ladder_demos.json	results/T11/no_ladder_demos.json; results/T11/definability_metrics.json	FIG-6:repo_asset; Figure 6 ([FIG-6]). Ladder front: fixed idempotent packaging saturate...
NG_LADDER_BOUNDED_INTERFACE	NG_LADDER_BOUNDED_INTERFACE.core	results/T11/definability_metrics.json	results/T11/definability_metrics.json; results/T11/no_ladder_demos.json	TBL-6:repo_asset; Table 6 ([TBL-6]). Ladder theorem evidence, count core, and corollary...

Continued on next page

Theorem	Claim	Primary handle	Evidence artifacts	Asset / callout decision
NG_ARROW_DPI	NG_ARROW_ DPI.guardrail	results/T15/adversarial_ cases.json	results/T15/adversarial_ cases.json; results/T15/initial_ distribution_sweep.csv	TBL-2:repo_asset; Table 2 ([TBL-2]). Arrow / protocol theorem evidence and guardrails: ...
NG_PROTOCOL_TRAP	NG_PROTOCOL_ TRAP.guardrail	results/T15/initial_ distribution_sweep.csv	results/T15/initial_ distribution_sweep.csv	TBL-2:repo_asset; Table 2 ([TBL-2]). Arrow / protocol theorem evidence and guardrails: ...
NG_FORCE_FOREST	NG_FORCE_ FOREST.guardrail	results/T15/threshold_ sweep.csv	results/T15/threshold_sweep.csv; results/T14/primitive_ coverage.json	TBL-3:repo_asset; Table 3 ([TBL-3]). Graph / affinity theorem evidence and guardrails: ...
NG_FORCE_NULL	NG_FORCE_ NULL.guardrail	results/T15/threshold_ sweep.csv	results/T15/threshold_sweep.csv; results/T14/primitive_ coverage.json	TBL-3:repo_asset; Table 3 ([TBL-3]). Graph / affinity theorem evidence and guardrails: ...
NG_MACRO_CLOSURE_ DEFICIT	NG_MACRO_CLOSURE_ DEFICIT.guardrail	results/T09/grid_ check.json	results/T09/grid_check.json; results/T09/closure_ metrics.json	TBL-4:repo_asset; Table 4 ([TBL-4]). Closure theorem evidence and provenance: the direc...
NG_OBJECT_ CONTRACTIVE	NG_OBJECT_ CONTRACTIVE.guardrail	results/T15/objecthood_ metric_sweep.csv	results/T15/objecthood_metric_ sweep.csv; results/T10/stability_grid.json	TBL-5:repo_asset; Table 5 ([TBL-5]). Objecthood admissible regime and guardrail matrix:...
NG_LADDER_ BOUNDED_INTERFACE	NG_LADDER_BOUNDED_ INTERFACE.guardrail	results/T11/no_ladder_ demos.json	results/T11/no_ladder_ demos.json; results/T11/definability_ metrics.json	FIG-6:repo_asset; Figure 6 ([FIG-6]). Ladder front: fixed idempotent packaging saturate...

Table 2: Frozen callout bindings and caption decisions.

Callout	Theorem ids	Primary source	Supporting refs	Caption / note
FIG-1	none	authored:theory-package- and-audit-convention- schematic	docs/project/theorem_ atlas.yaml; docs/project/claim_audit_ freeze.yaml	Figure 1 ([FIG-1]). Finite audited theory package and audit-convention schematic: microstate s...
FIG-2	NG_ARROW_DPI; NG_PROTOCOL_TRAP	results/T15/horizon_lens_ sweep.csv	results/T08/arrow_ metrics.json; results/T15/initial_ distribution_sweep.csv	Figure 2 ([FIG-2]). Hidden-clock DPI / protocol matrix: honest observed path-reversal KL can s...
FIG-3	NG_FORCE_FOREST; NG_FORCE_NULL	results/T14/primitive_ coverage.json	results/master/witness_ metrics.csv	Figure 3 ([FIG-3]). Graph / affinity contrasts: forest support removes cycle-space obstruction...

Continued on next page

Callout	Theorem ids	Primary source	Supporting refs	Caption / note
FIG-4	NG_MACRO_CLOSURE_ DEFICIT	results/T09/closure_ metrics.json	results/T09/grid_ check.json	Figure 4 ([FIG-4]). Closure front: exact packaged future laws define the finite variational ob. . .
FIG-5	NG_OBJECT_CONTRACTIVE	results/T10/objecthood_ metrics.json	results/T10/stability_ grid.json	Figure 5 ([FIG-5]). Objecthood front: in the admissible total-variation regime, contraction yi. . .
FIG-6	NG_LADDER_IDEM; NG_LADDER_BOUNDED_ INTERFACE	results/T11/no_ladder_ demos.json	results/T11/definability_ metrics.json	Figure 6 ([FIG-6]). Ladder front: fixed idempotent packaging saturates after one step, bounded. . .
TBL-1	NG_ARROW_DPI; NG_PROTOCOL_TRAP; NG_FORCE_FOREST; NG_FORCE_NULL; +4 more	results/master/witness_ metrics.csv	results/master/witness_ manifest.json	Table 1 ([TBL-1]). Theorem front map and support snapshot: the eight theorem fronts, their sec. . .
TBL-2	NG_ARROW_DPI; NG_PROTOCOL_TRAP	results/T15/adversarial_ cases.json	results/T15/initial_ distribution_sweep.csv	Table 2 ([TBL-2]). Arrow / protocol theorem evidence and guardrails: the paper binds the arrow. . .
TBL-3	NG_FORCE_FOREST; NG_FORCE_NULL	results/T15/threshold_ sweep.csv	results/T14/primitive_ coverage.json	Table 3 ([TBL-3]). Graph / affinity theorem evidence and guardrails: exact-support cycle infor. . .
TBL-4	NG_MACRO_CLOSURE_ DEFICIT	results/T09/grid_ check.json	results/T09/closure_ metrics.json	Table 4 ([TBL-4]). Closure theorem evidence and provenance: the direct pack uses exact variati. . .
TBL-5	NG_OBJECT_CONTRACTIVE	results/T15/objecthood_ metric_sweep.csv	results/T10/stability_ grid.json	Table 5 ([TBL-5]). Objecthood admissible regime and guardrail matrix: exact contraction metric. . .
TBL-6	NG_LADDER_IDEM; NG_LADDER_BOUNDED_ INTERFACE	results/T11/definability_ metrics.json	results/T11/no_ladder_ demos.json	Table 6 ([TBL-6]). Ladder theorem evidence, count core, and corollary map: no-ladder demos, ex. . .
TBL-A1	NG_ARROW_DPI; NG_PROTOCOL_TRAP; NG_FORCE_FOREST; NG_FORCE_NULL; +4 more	results/master/witness_ manifest.json	results/master/witness_ metrics.csv	Table A1 ([TBL-A1]). Artifact and evidence provenance map: each frozen core or guardrail claim. . .
TBL-A2	NG_ARROW_DPI; NG_PROTOCOL_TRAP; NG_FORCE_FOREST; NG_FORCE_NULL; +4 more	derived:lean-support-and- proof-dependency-map	docs/project/readiness_ checklist.yaml; docs/project/theorem_ atlas.yaml	Table A2 ([TBL-A2]). Lean support and proof dependency map: seven fronts are direct, objecthoo. . .

3 Computational audit and artifact manifest

The canonical command for rebuilding the computational side of the project is `make reproduce`. That command drives the reproducibility pipeline, refreshes theorem-pack summaries, Lean summaries, and rewrite-lane validations, and writes a manifest at `results/manifest/artifact_manifest.json`. The pipeline is deterministic up to the frozen repo contents and toolchain recorded in the manifest.

Table 3: Reproducibility manifest overview.

Field	Value
Canonical command	<code>make reproduce</code>
Manifest status	success
Generated at UTC	2026-03-22T07:25:15Z

Continued on next page

Field	Value
Total pipeline steps	62
Hashed files	350
Failed steps	0

The table below lists the theorem-pack and Lean-check steps that are directly relevant to theorem evidence in the main paper, together with the summary JSON outputs they refresh.

Table 4: Theorem-pack and Lean-check steps relevant to theorem evidence.

Step	Status	Command	Summary outputs
run_t15_robustness_suite	passed	python3 scripts/run_t15_robustness_suite.py	results/T15/fragility_summary.json
run_t22_dpi_protocol_pack	passed	python3 scripts/run_t22_dpi_protocol_pack.py	results/T22/dpi_protocol_pack_summary.json
run_t23_graph_affinity_pack	passed	python3 scripts/run_t23_graph_affinity_pack.py	results/T23/graph_affinity_pack_summary.json
run_t24_closure_pack	passed	python3 scripts/run_t24_closure_pack.py	results/T24/closure_pack_summary.json
run_t25_objecthood_pack	passed	python3 scripts/run_t25_objecthood_pack.py	results/T25/objecthood_pack_summary.json
run_t26_bounded_interface_pack	passed	python3 scripts/run_t26_bounded_interface_pack.py	results/T26/bounded_interface_pack_summary.json
run_t27_lean_graph_bridge	passed	python3 scripts/run_t27_lean_graph_bridge.py	results/T27/lean_graph_bridge_summary.json
run_t28_lean_finite_lens_count	passed	python3 scripts/run_t28_lean_finite_lens_count.py	results/T28/lean_finite_lens_count_summary.json
run_t29_lean_bounded_interface_corollary	passed	python3 scripts/run_t29_lean_bounded_interface_corollary.py	results/T29/lean_bounded_interface_corollary_summary.json
run_t30_lean_objecthood_uniqueness	passed	python3 scripts/run_t30_lean_objecthood_uniqueness.py	results/T30/lean_objecthood_uniqueness_summary.json
run_t31_paper_skeleton	passed	python3 scripts/run_t31_paper_skeleton.py	results/T31/paper_skeleton_summary.json
run_t32_asset_freeze	passed	python3 scripts/run_t32_asset_freeze.py	results/T32/asset_freeze_summary.json
run_t33_draft_v1	passed	python3 scripts/run_t33_draft_v1.py	results/T33/draft_v1_summary.json
run_t34_red_team_review	passed	python3 scripts/run_t34_red_team_review.py	results/T34/red_team_summary.json
run_t35_scope_charter	passed	python3 scripts/run_t35_scope_charter.py	results/T35/lean_probability_scope_charter_summary.json
run_t36_lean_probability_core	passed	python3 scripts/run_t36_lean_probability_core.py	results/T36/lean_probability_core_summary.json
run_t37_lean_kl_dpi_core	passed	python3 scripts/run_t37_lean_kl_dpi_core.py	results/T37/lean_kl_dpi_core_summary.json

Continued on next page

Step	Status	Command	Summary outputs
run_t38_lean_arrow_dpi	passed	python3 scripts/run_t38_lean_arrow_dpi.py	results/T38/lean_arrow_dpi_summary.json
run_t39_lean_protocol_trap	passed	python3 scripts/run_t39_lean_protocol_trap.py	results/T39/lean_protocol_trap_summary.json
run_t40_closure_feasibility	passed	python3 scripts/run_t40_closure_feasibility.py	results/T40/lean_closure_feasibility_summary.json
run_t41_lean_closure_variational_core	passed	python3 scripts/run_t41_lean_closure_variational_core.py	results/T41/lean_closure_variational_core_summary.json
run_t42_lean_closure_direct_pack	passed	python3 scripts/run_t42_lean_closure_direct_pack.py	results/T42/lean_closure_direct_pack_summary.json
run_t43_objecthood_tv_feasibility	passed	python3 scripts/run_t43_objecthood_tv_feasibility.py	results/T43/lean_objecthood_tv_feasibility_summary.json
run_t44_lean_objecthood_direct_pack	passed	python3 scripts/run_t44_lean_objecthood_direct_pack.py	results/T44/lean_objecthood_direct_pack_summary.json

4 Fragility diagnostics

The main paper states theorem objects in exact finite terms. The fragility layer records where nearby proxy choices can distort interpretation without changing the underlying theorem statements. The frozen source for this section is `results/T15/fragility_summary.json`, together with the guardrail claims in `docs/project/claim_audit_freeze.yaml`.

The dominant risks are: changing the initial law in stationary arrow claims, changing the observation lens, fitting first-order macro proxies, thresholding exact support graphs, and switching to non-admissible objecthood metrics. Each such move is a scope change or a proxy diagnostic, not an in-theorem modification of the audited object.

Table 5: Fragility items and their frozen guardrails.

Fragility item	Severity	Affected claims	Guardrail claim	Mitigation / note
stationary_zero_arrow_is_fragile_to_initial_distribution	high	NG_PROTOCOL_TRAP.core	NG_PROTOCOL_TRAP.guardrail	Protocol-trap claims must be stated on the lifted chain or honest observed path law, under sta...
macro_arrow_depends_strongly_on_lens_choice	high	NG_ARROW_DPI.core; NG_PROTOCOL_TRAP.core	NG_ARROW_DPI.guardrail	Fitted first-order macro proxies are not theorem-level arrow audits and can create or distort ...
fitted_markov_proxy_can_create_or_distort_macro_arrow	high	NG_ARROW_DPI.guardrail; NG_PROTOCOL_TRAP.guardrail; NG_MACRO_CLOSURE_DEFICIT.guardrail	NG_ARROW_DPI.guardrail	Fitted first-order macro proxies are not theorem-level arrow audits and can create or distort ...

Continued on next page

Fragility item	Severity	Affected claims	Guardrail claim	Mitigation / note
threshold_floors_can_introduce_spurious_one_way_support	medium	NG_FORCE_FOREST.guardrail; NG_FORCE_NULL.guardrail	NG_FORCE_NULL.guardrail	Thresholding or regularization can distort exactness and affinity interpretation; theorem clai...
object_multiplicity_can_be_metric_artifact	medium	NG_OBJECT_CONTRACTIVE.guardrail	NG_OBJECT_CONTRACTIVE.guardrail	Theorem objecthood claims must use admissible total-variation/Dobrushin metric assumptions; pr...

5 Formalization support map

All eight theorem fronts are formalized directly in Lean 4 (toolchain v4.28.0, Mathlib pinned to the matching release). The current coverage receipt is `review/current-theorem-coverage.json`; it records the exported declarations, their precise scope, source hashes, and the results of a clean root build, explicit target-type checks (`review/CheckTargets.lean`), and a transitive axiom audit (`review/CheckAxioms.lean`). The declarations below depend only on `propext`, `Classical.choice`, and `Quot.sound`. The receipt is regenerated by `python3 scripts/run_mathematical_audit.py`, and the semantic review of each formal statement is in `review/mathematical-review.txt`.

Table 6: Lean support across theorem fronts.

Theorem	Lean modules	Main declarations	Scope
NG_ARROW_DPI	RealFiniteKL; RealArrowDPI;	RealProbability.markovArrowDPI;	Any initial law, kernel, lens, horizon;
NG_PROTOCOL_TRAP	RealMarkovPaths	RealProbability.KL_pushforward_le	extended-real KL.
NG_FORCE_FOREST	RealMarkovPaths	RealProbability.protocolTrap;	Detailed balance only; null states allowed;
		RealProbability.detailedBalance_pathLaw_reversible	reversal symmetry derived.
	RealGraphExactness	RealGraph.forest_exact	Any acyclic simple graph; identity on edges only.
NG_FORCE_NULL	RealGraphExactness	RealGraph.exact_closedWalk_zero	Edge-local gradient; every closed walk.
NG_MACRO_CLOSURE_DEFICIT	RealKLMixture; RealClosure	RealProbability.closureTheorem;	Arbitrary current law at time zero; any lag;
		RealProbability.closure_variational_minimum;	all stochastic candidate kernels.
		RealProbability.conditionalInfo_positive_of_distinct_fiber_rows	
NG_OBJECT_CONTRACTIVE	RealTVContraction	RealProbability.dobrushin_contraction;	Nonempty finite space; full $2\varepsilon/(1 - \lambda)$ bound and uniqueness.
		RealProbability.contractive_separation;	
		RealProbability.contractive_stationary_unique	
NG_LADDER_IDEM	Idempotence	iterate_stabilizes_ext	Any carrier.
NG_LADDER_BOUNDED_INTERFACE	FiniteImageDefinability	FiniteImage.definable_cardinality;	Arbitrary domain and codomain with finite image.
		FiniteImage.no_infinite_distinct_sequence	

Existence of a stationary law and geometric convergence under strict contraction are proved analytically in the main paper and are not exported as Lean declarations. The older per-front bridge files under `docs/project/lean_*.yaml` and the readiness checklist are retained as historical provenance; their earlier status labels are not proof certificates and are superseded by the coverage receipt.